

MORGAN STANLEY FTE INTERVIEW EXPERIENCE

ABOUT THE COMPANY:

Morgan Stanley operates as a global financial services company. It mobilizes capital to help government, corporations, institutions and individuals around the world achieve their financial goals. Morgan Stanley is consistently highly ranked in mergers and acquisitions and initial public offerings.



Eligibility Criteria: CGPA - 8.0 and above

Branches Considered: CS & IT

CTC: 29.48 LPA

Sequence of Selection: PPT & Online Assessment, 2 Technical Interviews and HR Interview.

Job Role Offered: Technology Analyst

ROUND 1 – ONLINE ASSESSMENT

Duration: 90 minutes (about 1 and a half hours)

The assessment was conducted through SHL platform. It consisted of three sections: Aptitude, Debugging and Coding.

Aptitude – This section was of medium-hard difficulty level.

Debugging – This section consisted of code snippets to rectify the logic. This was of easy-medium difficulty level. Initially chosen programming language cannot be changed till the end.

Coding – This section consisted of 3 coding questions. The three questions were easy. As the questions were easy, the time of submission played a greater role in selection.

Every section was time bounded, that is a section cannot be backtracked once its time slice is over.

After the online assessment, 14 members were shortlisted for the next round – Technical Interview I.

ROUND 2 – TECHNICAL INTERVIEW I

Mode: Physical

Duration: 1 hour

Difficulty level: Easy-Medium

The interview was based on OS, SQL, DSA, OOPS. It almost covered all major CS domains. The interviewer was friendly and he made me feel comfortable so that my anxiety and nervousness came down. The interviewer started off with some casual questions like where do I come from? And then, he asked me what is the eligibility criterion to enter Anna University? Then he started asking technical questions. Below are the questions asked:

I was asked to debug a code snippet which was written in Cpp . It was a code snippet which included dynamic memory allocation using new pointer, array declaration and stuff like that. I was able to debug it completely.

What is the use of header files? Why do we use it?

Question on cpp library - vector.h. (I am good at Java language. So, I said to the interviewer that I am good and comfortable with Java. So, he started asking Java questions.)

Questions on OOPS:

Define an abstract class. Write the code to illustrate an example of an abstract class.

What are the access specifiers available in Java Programming language?

Questions on OS:

What are threads?

What is a deadlock?

What are the necessary conditions for a deadlock to occur?

Questions on SQL:

Write a query to create a table.

Write a query to insert data into a table.

What is left outer join, inner join and cross join?

Coding questions:

What is time complexity? Write the time complexity classes in ascending order.

Write any sorting algorithm. Explain the working of it. Write down the best-case time complexity for it.

What is the difference between a run-time error and a compile time-error?

Given four persons at designations – D1, D2, D3, D4, respectively where D1 is the least designation and D4 is the highest. The person can skip one position and get promoted to the second next higher position or he can get promoted to the next higher position. Write down the possible ways to reach D4. This question is similar to this problem. STAIRCASE PROBLEM - <https://www.geeksforgeeks.org/count-ways-reach-nth-stair/>

Given login time, logout time for n employees in a company. The task is to find the minimum number of chairs in a company to accommodate 'n' employees. This question is like this problem – MINIMUM NUMBER OF PLATFORMS REQUIRED FOR A RAILWAY/BUS STATION - <https://www.geeksforgeeks.org/minimum-number-platforms-required-railwaybus-station/>

Write a program to find if a string is palindrome or not.

For the coding questions, I was asked to write the logic.

I was able to answer all the questions. The most important thing is to communicate with the interviewer throughout the interview session. Even if you get stuck up thinking logic, the interviewer gives you hint to crack it. So, communicate your thoughts throughout. Do not hesitate to do that assuming that you are wrong. Be bold and confident with your answers.

Shortlisting was done for the next round.

ROUND 3 – TECHNICAL INTERVIEW II

Mode: Physical

Duration: 45 minutes

Difficulty level: Easy

The interviewer started off asking me to give a self intro. This round was on System Design. He gave me a scenario to design a system. I was asked to design a vending system which is used in malls.

Functionalities required:

- Display the items available.
- Receive the order.
- Check the inventory.
- Check for successful payment and deliver the products.
- Update the inventory.

I was asked to draw the class diagram for the system proposed. Usage of appropriate data structures in some of the functionalities were asked. I made corrections as per his views. He asked me to draw the database tables required for this system. I included a third-party Payment Processing system in my diagram. So, he asked me to draw a class diagram which focuses on the Payment modes – UPI, Cash and Card. He asked me to explain the OOPS concept involved in that diagram. Some OS questions which were asked:

What is multithreading? What are its benefits? What are the ways in which multithreading can be established using Java?

Write a code snippet which illustrates the principle of Mutual Exclusion.

Which is the keyword in Java used to ensure mutual exclusion?

Given two tables – Students information table and Company selects table. Write a SQL query to find the name of the student and the company name where he got placed who belongs to XYZ college and receives the highest salary offered in the city Mumbai.

Questions were asked on the tech stack mentioned in the projects.

Finally, I was asked to ask a question to the interviewer. I asked him about the work culture at Morgan and being a FinTech company, how does it differ from the other product-based companies.

I found him impressed with my answers and had a hope that I would get selected for the next round. Shortlisting was done for the final round.

ROUND 4 – HR INTERVIEW

Mode: Physical

Duration: 25 minutes

This round seemed little harder than the previous rounds. The interviewer greeted me and started asking the questions. The questions were like:

How many projects have you worked on?

What are the difficulties that you have faced while working on projects?

The requirements of the project changed in the middle while your project was half way completed. How would handle that circumstance?

Tell me about the toughest decision that you have taken while working on projects.

Assume that you are the team lead and your teammate is not very co-operative. He is very lethargic and does not submit the assigned work within the deadline. What will be your action towards him as a team lead?

Assume that you are proposing a solution to a problem while the other person is proposing an efficient one compared to yours. Will you give up on your idea or stand by yours?

Tell me your experience when you have taken risks while working on projects and the results occurred from your risks were not as you expected.

I found this round a very formal discussion and it was a bit tough. Going through sample HR questions and a light preparation, rather than on spot answering will help you ace the HR interview.

3 of us were selected as FTE at Morgan Stanley.

Tips:

Strictly restrict your resume to 1 page. List out honestly only the skills what you possess in your resume.

Think aloud. Express your views and ideas.

Be strong in OOPS concepts, DBMS, OS, DSA.

Before the interview, read about the company well.

All the very best!

PRIYADHARSHINI P (2019-2023)

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